Systems

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Systems

- Static and dynamic behavior
- Models
- Proportional systems
- First-order systems
- Second-order systems
- Linear Time-Invariant systems
- Dead time

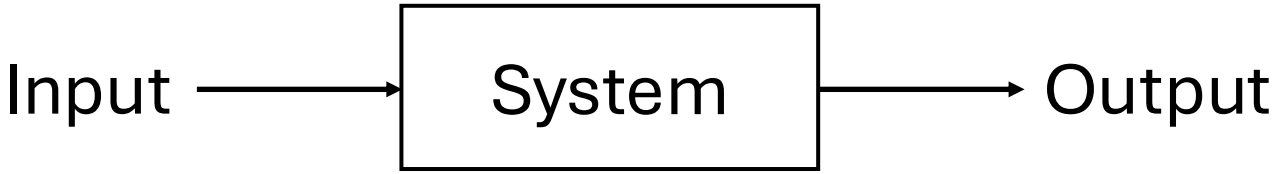
Processes, Systems, Models

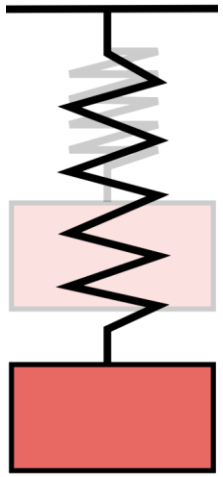
Process: the real-world phenomenon (physical, biological, ...), aka plant

System: abstraction of that process (behavior, dynamics)

Model: mathematical representation of the system

A system maps inputs to outputs (just like processes)

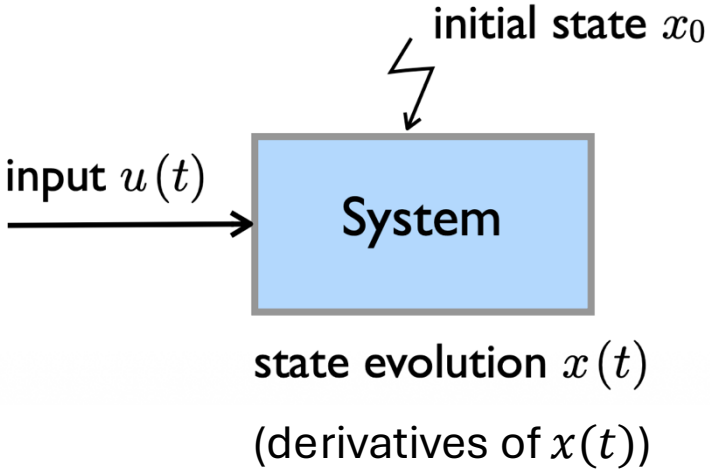




input: external force

output: displacement

Static and Dynamic Systems



Static system:

Output depends only on the current input, no memory

Time-varying ≠ dynamic

Examples:

- Resistor (model: $v = R \cdot i$)
- Valve

Dynamic system:

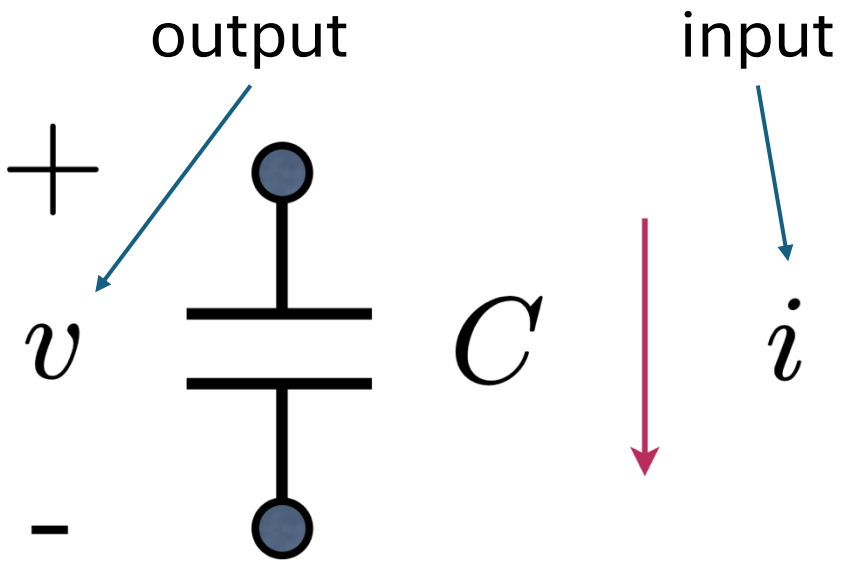
Output depends on input and past states → evolves over time

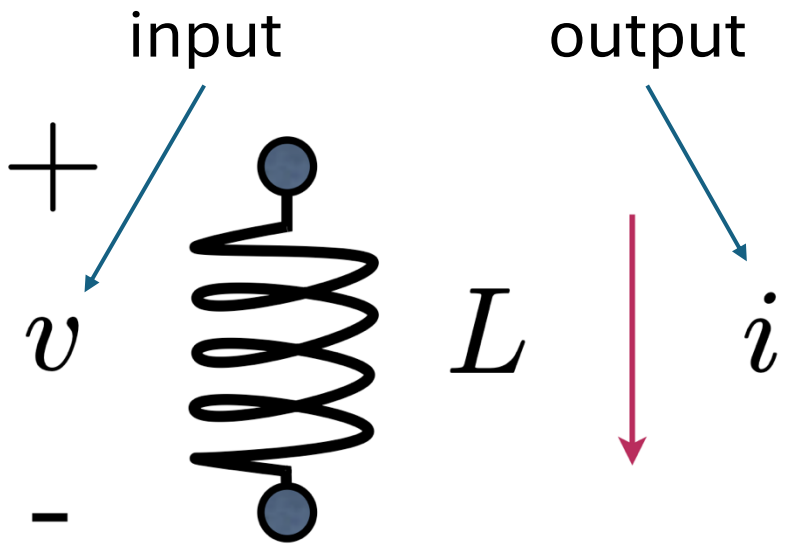
Energy storage → memory

Examples:

- Capacitor
- Mass–spring–damper

Examples from Electrical Engineering





$$\frac{dv}{dt} = \frac{1}{C} i \quad \xleftarrow{\text{model}} \quad \frac{di}{dt} = \frac{1}{L} v$$

Mathematical Models

$$\frac{dx}{dt} = \dot{x}$$

- Algebraic equations (static systems)

$$y(t) = f(u(t), t)$$

output

input

- Differential equations (dynamic systems)

$$\dot{x}(t) = f(x(t), u(t), t)$$

state variable

- Difference equations (discrete-time systems)

$$x(t+1) = f(x(t), u(t), t)$$

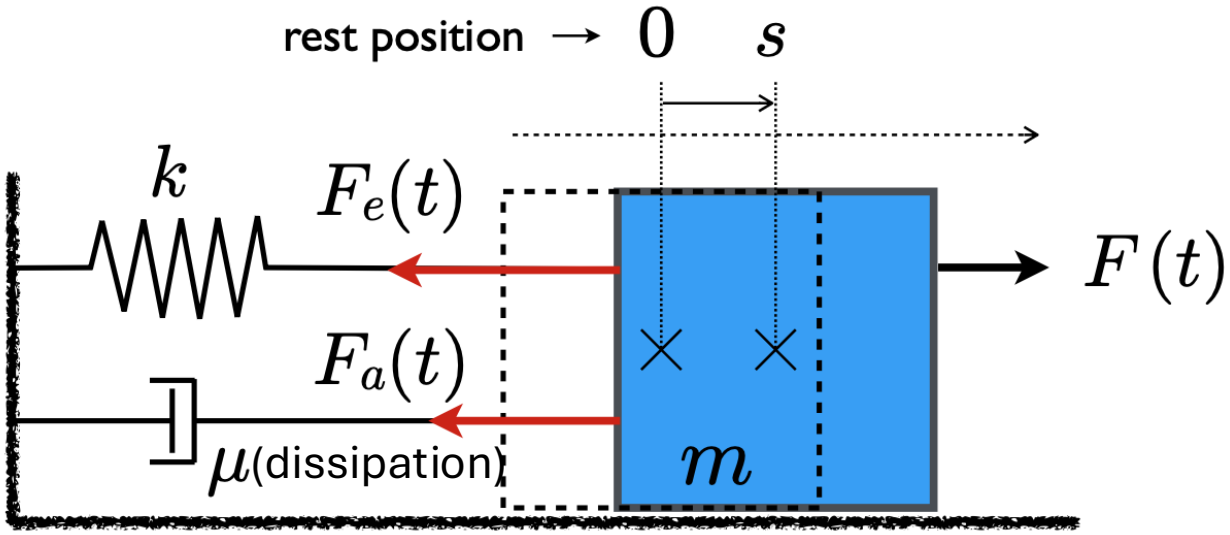
Define output separately:

$$y = x$$
$$y = \dot{x}$$
$$\ldots$$

Differential-Equation Models

Describe dynamics using derivatives

Example: mass–spring–damper system



Newton:

$$ma = F(t) - F_e(t) - F_a(t)$$

→ Model:

$$m\ddot{x} + \mu\dot{x} + kx = F(t)$$

Order of equation = system order

- = number of state variables
- = number of energy storage elements

second-order system: energy storages mass and spring

(even if we choose $y = \dot{x}$, first-order only if we removed the spring)

Why We Need Models

- Predict system behavior
- Design controllers before implementation
- Understand cause-and-effect relationships
- Simplify complex real-world systems

White-, Gray-, and Black-Box Models

	White-Box	**Gray-Box**	**Black-Box**
System Knowledge	Full	Partial	None
Parameter Source	Physical	Mixed	Data Only
Interpretability	High	Medium	Low
Effort / Complexity	High	Medium	Low-Medium
Extrapolation	Reliable	Moderate	Poor

Proportional (P) Systems

- Output proportional to input

$$y(t) = K \cdot u(t)$$

- Static gain (gain = output divided by input), aka DC gain (zero frequency)

- No dynamics

- Examples:
  - Amplifier
  - Resistor: $v(t) = R \cdot i(t)$

Integrating and Differentiating Behavior

- Integrator: $y(t) = \int u(t)\, dt$
  - Accumulates input over time
  - Introduces memory (energy storage)

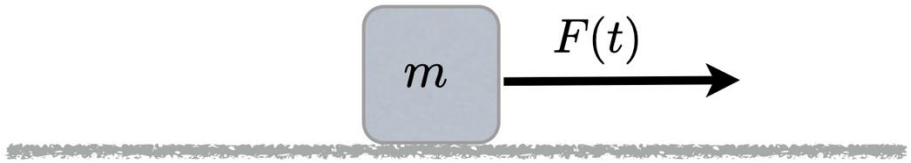
- Differentiator: $y(t) = \frac{d}{dt} u(t)$
  - Responds to rate of change
  - Sensitive to fast variations

dynamic

Physical systems often contain integrators and differentiators implicitly

→ dynamic behavior (PT1, PT2, ...)

Example: Free Fall



Velocity of a mass subject to a force: $m \cdot \dot{v} = F(t)$

output input

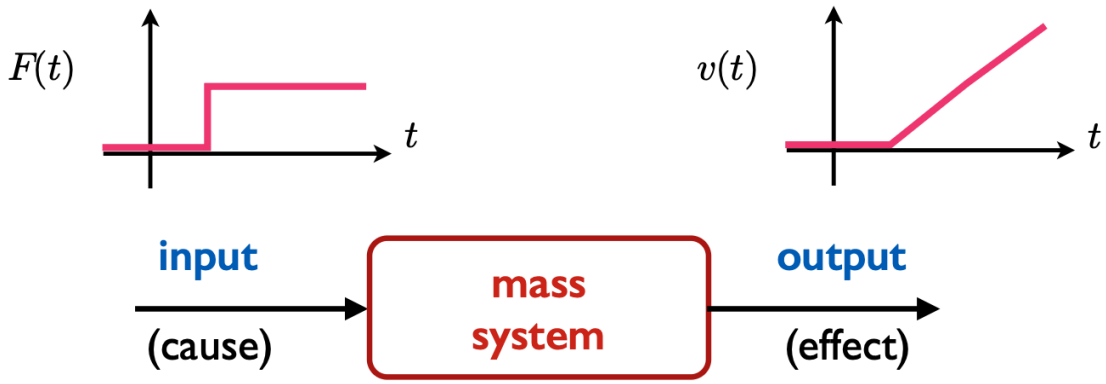
Freefall: $\dot{v} = g$

earth‘s acceleration (gravity)

Starting at $t_0 = 0$ with $v_0 = 0$:

$$v = \int_0^t g \, dt = g \cdot t$$

→ grows linearly over time



Real-world: damping due to air resistance …

First-Order Systems (PT1)

A sudden change of the input signal to a new constant value, maintained thereafter:

$$u(t) = \begin{cases} 0, & t < 0 \\ u_{\text{step}}, & t \geq 0 \end{cases}$$

One energy storage element (→ 1)

$$\tau \cdot \dot{y}(t) + y(t) = K \cdot u(t)$$

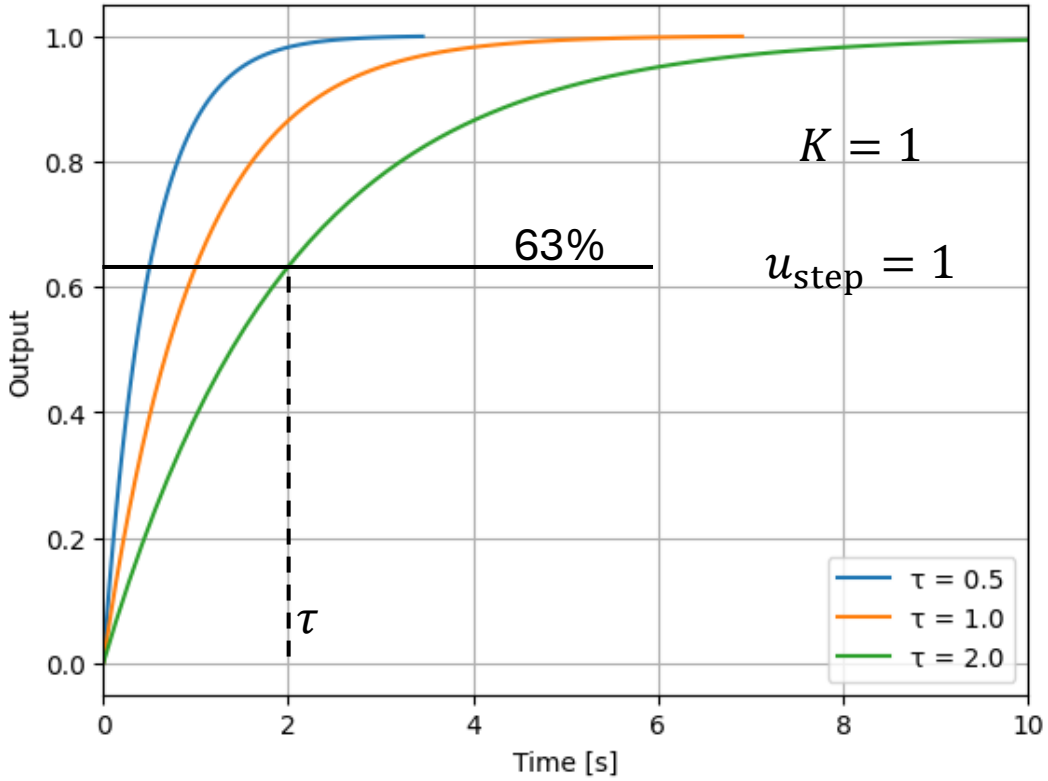
Two system parameters:

- Time constant τ → dynamic (T)
- Static (DC) gain K → proportional (P)

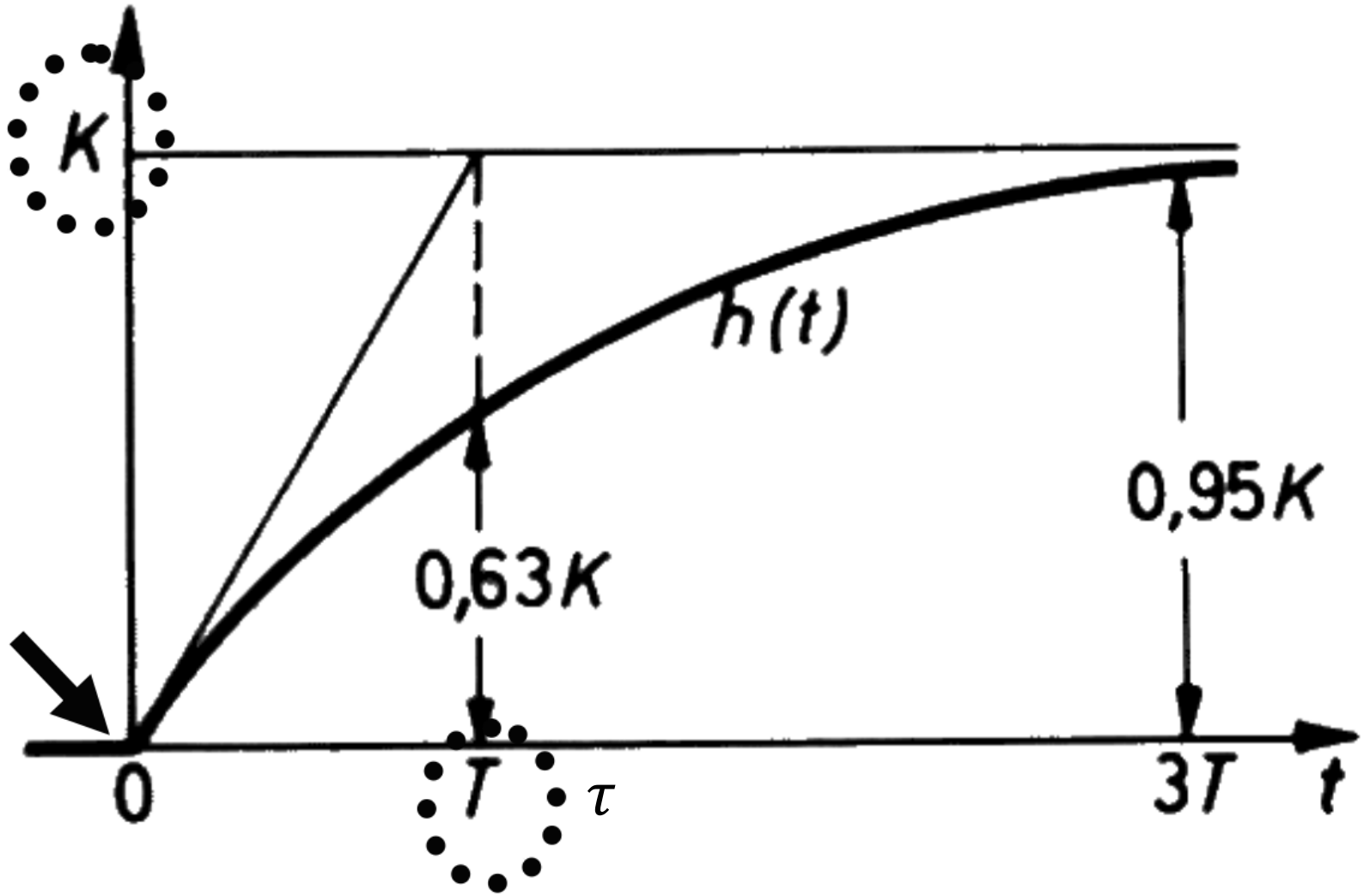
$$K = \frac{y(\infty)}{u_{\text{step}}}$$

output–input ratio for a constant input after transients have died out, i.e., $\dot{y} = 0$

Step response:



Gain and Time Constant



Solution for PT1 Differential Equation

Unforced (natural): $\tau \cdot \dot{y}(t) + y(t) = 0$

$$y(t) = C \cdot e^{-t/\tau}$$

defined by initial conditions

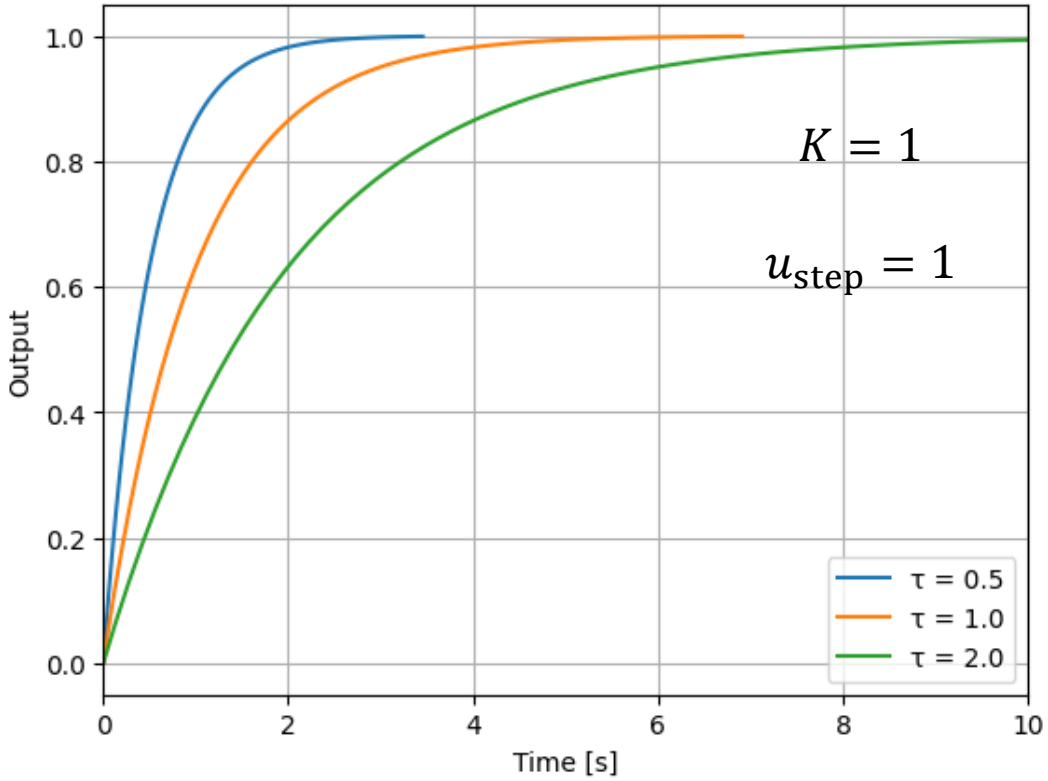
Forced: $\tau \cdot \dot{y}(t) + y(t) = K \cdot u(t)$

Example:

- At $t = 0$, step from $u = 0$ to $u = u_{\text{step}}$
- Initial condition $y(0) = 0$

$$y(t) = K \cdot u_{\text{step}} \cdot \left(1 - e^{-t/\tau}\right)$$

Step response:



Derivation of Specific Example

Total response = forced (particular) response + natural (homogeneous) response

$$y(t) = y_p + C \cdot e^{-t/\tau}$$

Particular response: constant input (u_{step}) → constant solution y_p

$$0 + y_p = K \cdot u_{\text{step}}$$

$$\Rightarrow y(t) = K \cdot u_{\text{step}} + C \cdot e^{-t/\tau}$$

Determine C from initial condition ($y(0) = 0$)

$$0 = K \cdot u_{\text{step}} + C \cdot e^{-0/\tau}$$

$$\Rightarrow C = -K \cdot u_{\text{step}}$$

$$\Rightarrow y(t) = K \cdot u_{\text{step}} \cdot \left(1 - e^{-t/\tau}\right)$$

Free Fall with Viscous Damping

Account for air resistance:

$$m \cdot \dot{v} = m \cdot g - c \cdot v$$

damping coefficient

→ PT1 differential equation:

$$\frac{m}{c} \cdot \dot{v} + v = \frac{m}{c} \cdot g$$

$$\tau = \frac{m}{c} \qquad K = \frac{m}{c}$$

Viscous damping is a *simplified model*.
In reality, aerodynamic drag is typically quadratic in velocity.
Linear damping is a good approximation for small velocities or near an operating point.

Example: RC and RL Circuits

Transient responses to switching

Capacitor: $\frac{dv}{dt} = \frac{1}{C} i$

Inductor: $\frac{di}{dt} = \frac{1}{L} v$

But circuits without R are not physically realizable (e.g., charging a capacitor would require infinite current)

→ damping

Transient Responses: RC Series Circuit

Differential equation:

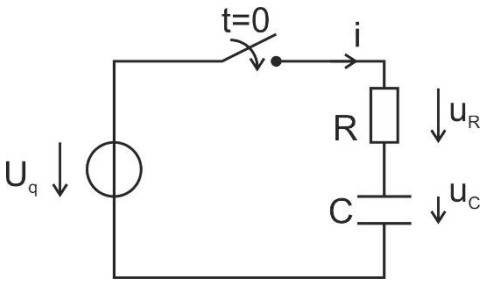
$$R \cdot i + u_C = U_q$$

$$\rightarrow \underbrace{R \cdot C}_{} \cdot \dot{u}_c + u_C = U_q$$

$$\tau = RC \qquad K = 1$$

system parameters

Charging the capacitor with DC:

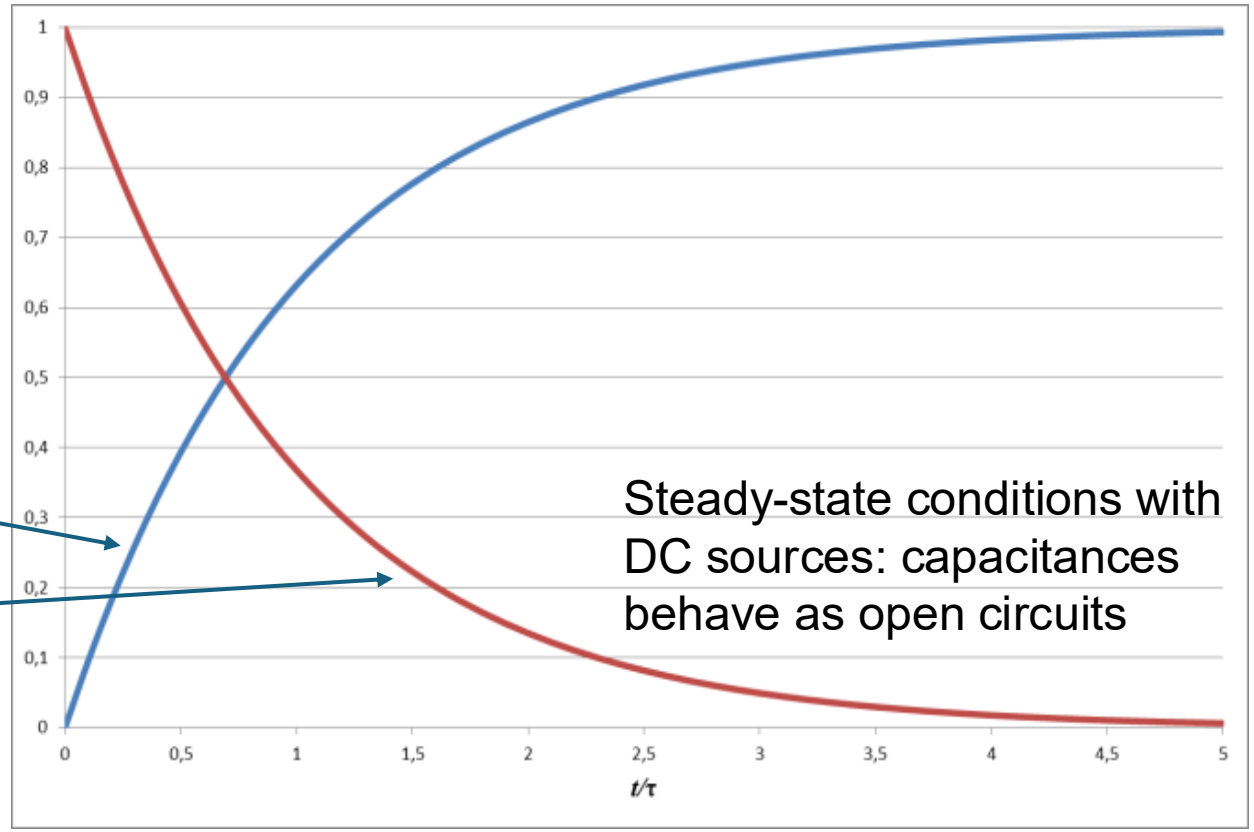


$$u_C = U_q\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$i = C \cdot \dot{u}_c = \frac{U_q}{R} \cdot e^{-\frac{t}{\tau}}$$

$$u_R = R \cdot i = U_q \cdot e^{-\frac{t}{\tau}}$$

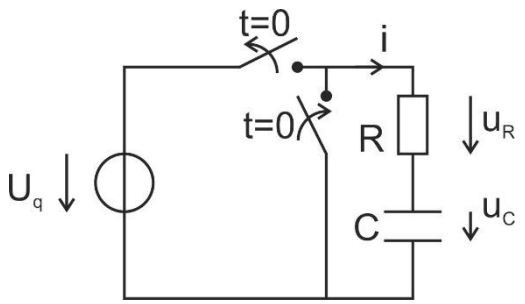
Transients: time-varying currents and voltages resulting from the sudden application of sources (usually due to switching)



Steady-state conditions with DC sources: capacitances behave as open circuits

Transient Responses: RC Series Circuit

Discharging the capacitor:



Differential equation:

$$R \cdot C \cdot \dot{u}_c + u_C = 0$$

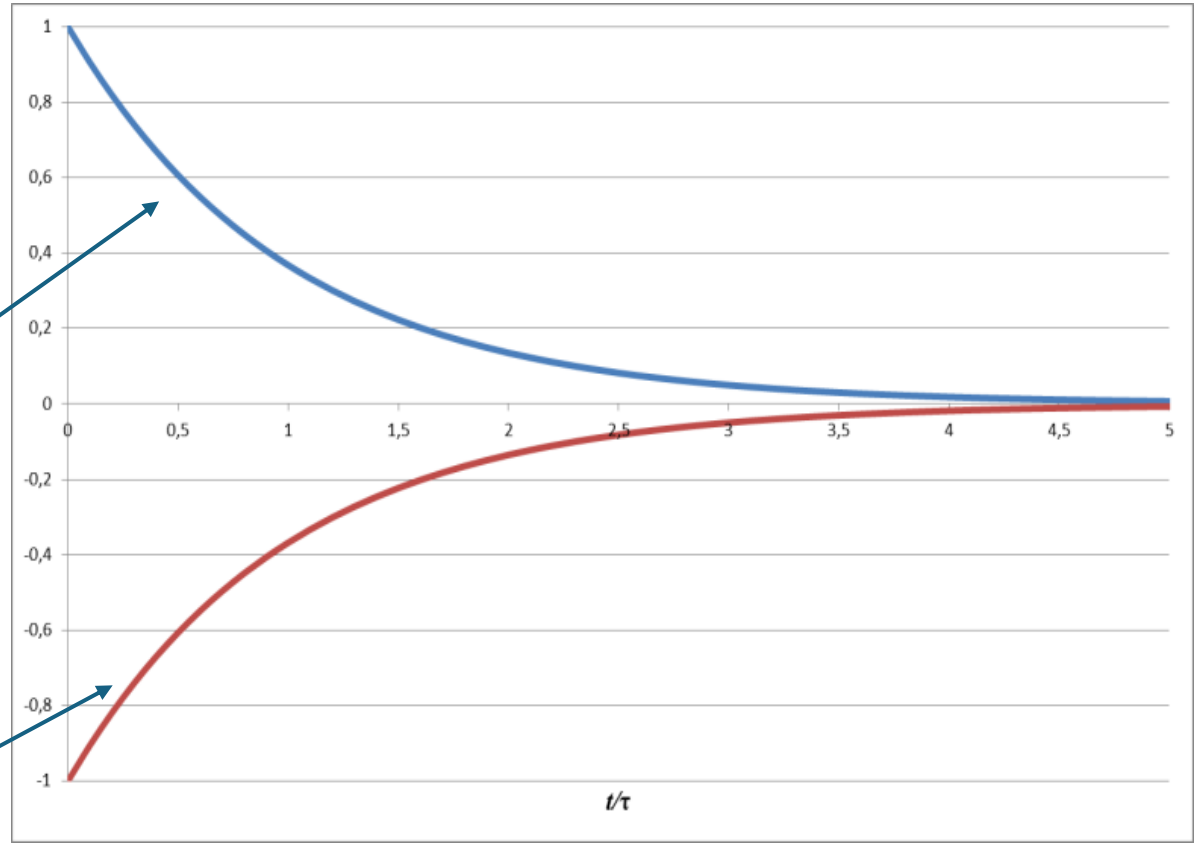
$$\tau = RC$$

$$u_C = U_q \cdot e^{-\frac{t}{\tau}}$$

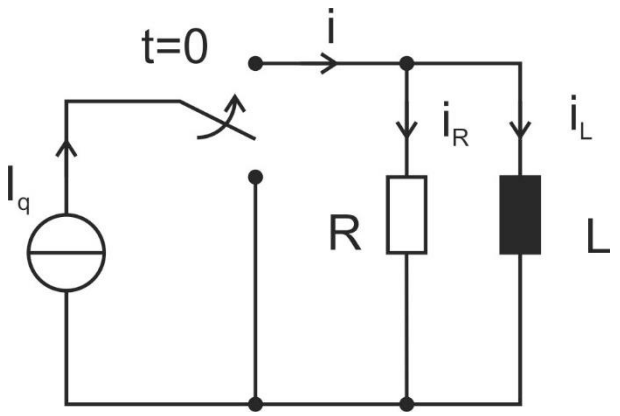
$u_C(0)$

$$i = -\frac{U_q}{R} \cdot e^{-\frac{t}{\tau}}$$

$C \cdot \dot{u}_c$



Transient Responses: RL Parallel Circuit



$$i_R + i_L = I_q$$

$$\rightarrow \frac{u}{R} + i_L = I_q$$

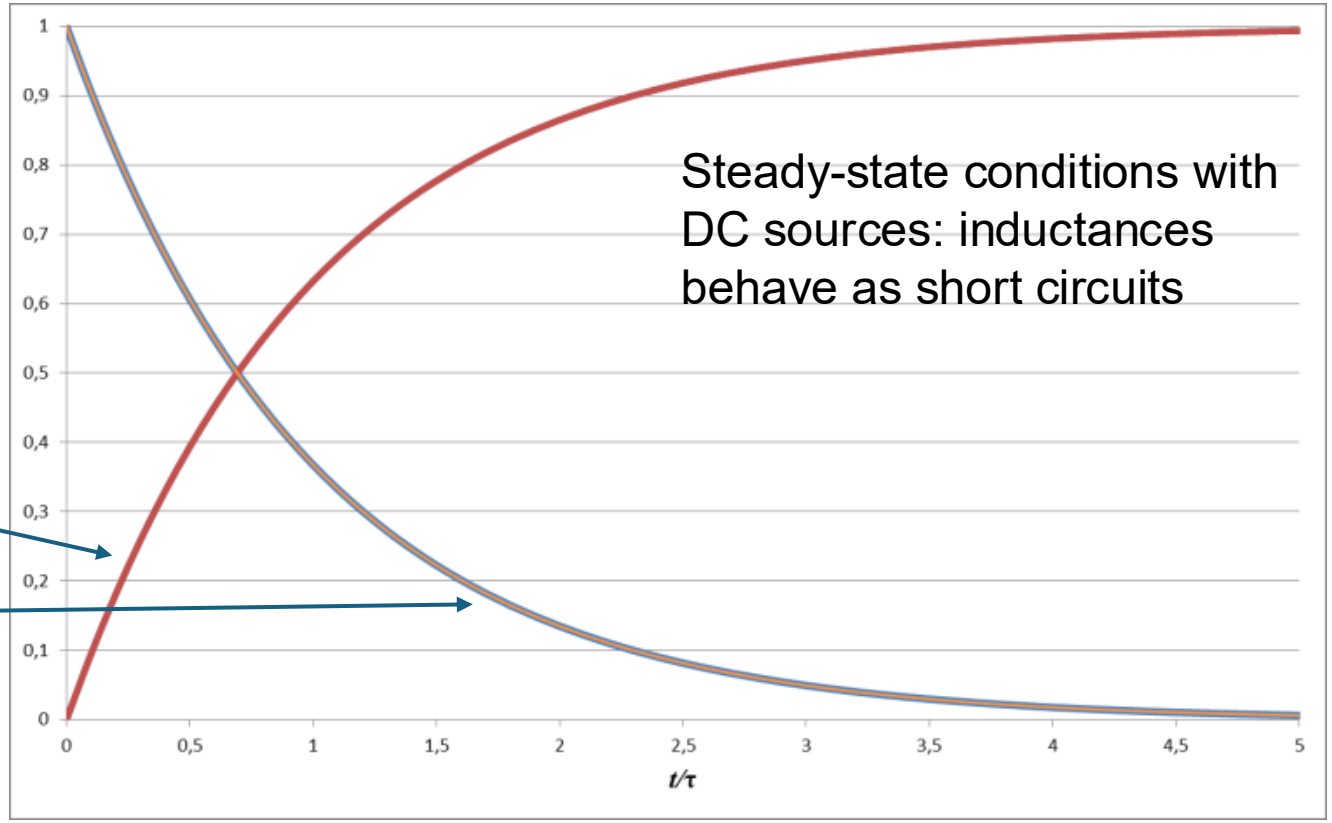
$$\rightarrow \frac{L}{R} \cdot \frac{di_L}{dt} + i_L = I_q$$

$$\tau = \frac{L}{R}$$

$$i_L = I_q\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$u = L \cdot \frac{di_L}{dt} = I_q \cdot R \cdot e^{-\frac{t}{\tau}}$$

$$i_R = I_q - i_L = I_q \cdot e^{-\frac{t}{\tau}}$$



Series connection of two PT1:
Time constants determine behavior

Second-Order Systems (PT2)

Two energy storage elements (→ 2)

$$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 K u$$

Three system parameters:

- Natural frequency ω_n
- Damping ratio ζ
- Static (DC) gain (as in PT1): $K = \frac{y(\infty)}{u_{\text{step}}}$

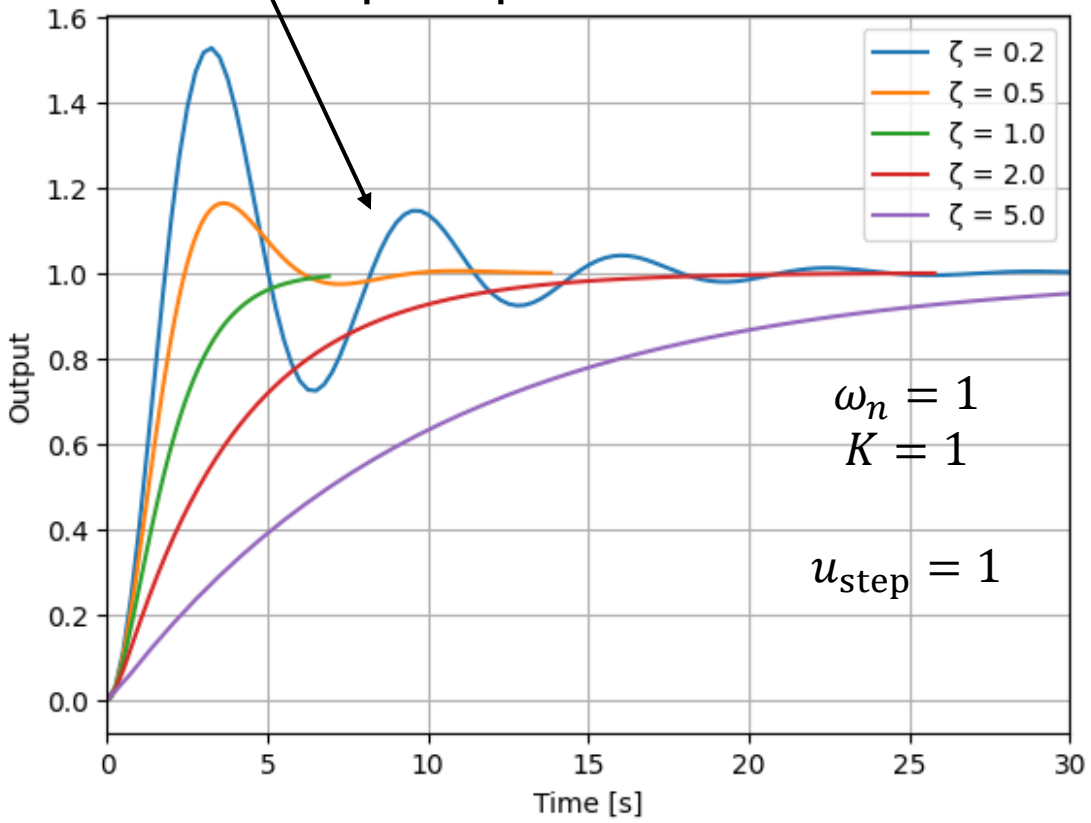
Different cases:

- Underdamped ($\zeta < 1$)
- Critically damped ($\zeta = 1$)
- Overdamped ($\zeta > 1$)

} no oscillation

can show oscillations

Step response:



Solution for PT2 Differential Equation

Unforced (natural): $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = 0$

$\zeta < 1$: $y(t) = e^{-\zeta\omega_n t}\left(C_1 \cos\left(\sqrt{1-\zeta^2}\omega_n t\right) + C_2 \sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right)$

$\zeta = 1$: $y(t) = (C_1 + C_2 t)e^{-\omega_n t}$

$\zeta > 1$: $y(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$ with $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

Forced: $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 K u$

Example:

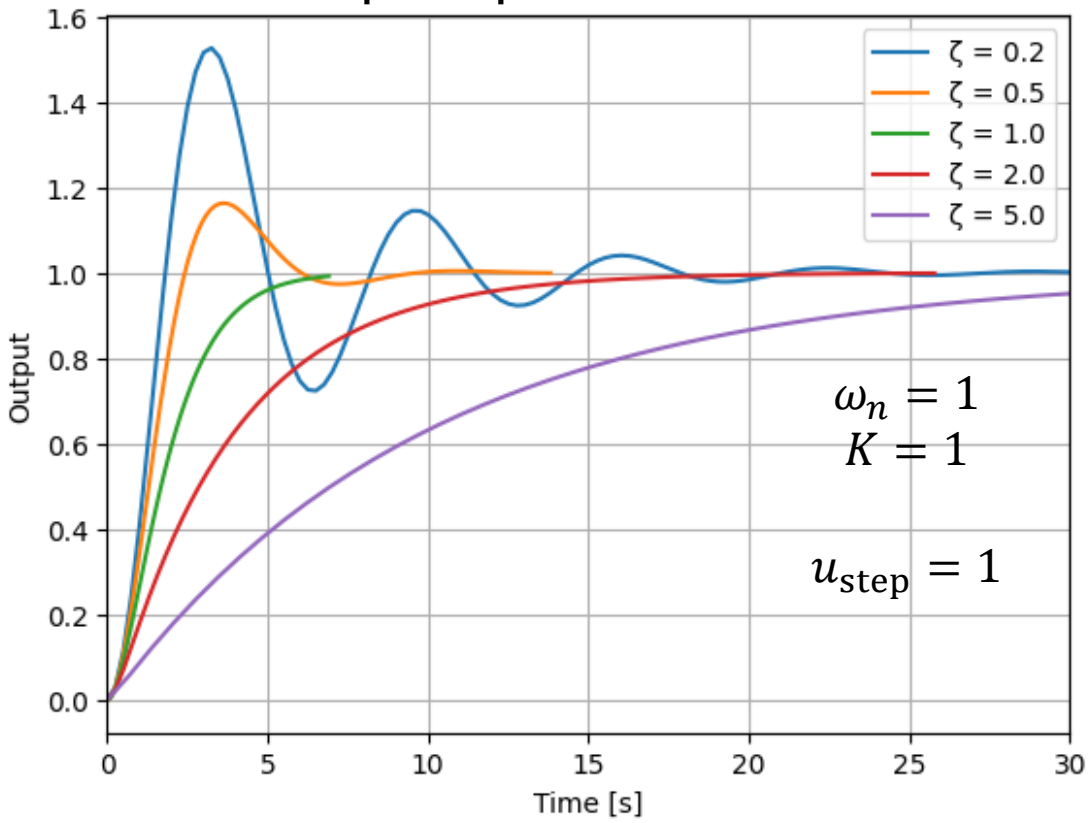
- At $t = 0$, step from $u = 0$ to $u = 1$
- Initial conditions $y(0) = 0, \dot{y}(0) = 0$

$\zeta < 1$: $y(t) = K\left[1 - e^{-\zeta\omega_n t}\left(\cos\left(\sqrt{1-\zeta^2}\omega_n t\right) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right)\right]$

$\zeta = 1$: $y(t) = K[1 - (1 + \omega_n t)e^{-\omega_n t}]$

$\zeta > 1$: $y(t) = K\left[1 + \frac{s_2}{s_1 - s_2}e^{s_1 t} - \frac{s_1}{s_1 - s_2}e^{s_2 t}\right]$

Step response:



Derivation of Specific Example

Particular response: constant input ($u_{\text{step}} = 1$) → constant solution y_p

$$0 + 0 + \omega_n^2 y_p = \omega_n^2 \cdot K \cdot 1$$

$$\Rightarrow y_p = K$$

$\zeta < 1$:

Total response = forced (particular) response + natural (homogeneous) response

$$y(t) = K + e^{-\zeta\omega_n t}\left(C_1 \cos\left(\sqrt{1-\zeta^2}\omega_n t\right) + C_2 \sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right)$$

Determine C_1 and C_2 from initial conditions ($y(0) = 0, \dot{y}(0) = 0$)

$$0 = K + 1 \cdot (C_1 \cdot 1 + C_2 \cdot 0) \quad \Rightarrow C_1 = -K$$

$$\dot{y}(t) = e^{-\zeta\omega_n t}\left[-\zeta\omega_n(C_1\cos + C_2\sin) - \sqrt{1-\zeta^2}\omega_n C_1 \sin + \sqrt{1-\zeta^2}\omega_n C_2 \cos\right]$$

$$0 = 1 \cdot \left[-\zeta\omega_n(C_1 + 0) - 0 + \sqrt{1-\zeta^2}\omega_n C_2\right] \quad \Rightarrow C_2 = \frac{\zeta\omega_n C_1}{\sqrt{1-\zeta^2}\omega_n} = -\frac{\zeta K}{\sqrt{1-\zeta^2}}$$

$$\Rightarrow y(t) = K + e^{-\zeta\omega_n t}\left(-K\cos\left(\sqrt{1-\zeta^2}\omega_n t\right) - \frac{\zeta K}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right) =$$

$$K\left[1 - e^{-\zeta\omega_n t}\left(\cos\left(\sqrt{1-\zeta^2}\omega_n t\right) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right)\right]$$

$\zeta = 1$:

Total response = forced (particular) response + natural (homogeneous) response

$$y(t) = K + (C_1 + C_2 t)e^{-\omega_n t}$$

Determine C_1 and C_2 from initial conditions ($y(0) = 0, \dot{y}(0) = 0$)

$$0 = K + (C_1 + C_2 \cdot 0) \cdot 1 \quad \Rightarrow C_1 = -K$$

$$\dot{y}(t) = -C_1 \cdot e^{-\omega_n t} \cdot \omega_n - C_2 \cdot e^{-\omega_n t} \cdot \omega_n \cdot t + C_2 \cdot e^{-\omega_n t}$$

$$0 = -C_1 \cdot \omega_n - 0 + C_2 \quad \Rightarrow C_2 = C_1 \cdot \omega_n = -K \cdot \omega_n$$

$$\Rightarrow y(t) = K + (-K - K\omega_n t)e^{-\omega_n t} = K[1 - (1 + \omega_n t)e^{-\omega_n t}]$$

$\zeta > 1$:

Total response = forced (particular) response + natural (homogeneous) response

$$y(t) = K + C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

Determine C_1 and C_2 from initial conditions ($y(0) = 0, \dot{y}(0) = 0$)

$$0 = K + C_1 + C_2$$

$$\dot{y}(t) = s_1 \cdot C_1 e^{s_1 t} + s_2 \cdot C_2 e^{s_2 t}$$

$$0 = s_1 \cdot C_1 + s_2 \cdot C_2$$

$$\Rightarrow C_1 = -\frac{s_2 \cdot C_2}{s_1} = \frac{s_2 \cdot K + s_2 \cdot C_1}{s_1} \quad \Rightarrow C_1 - \frac{s_2 \cdot C_1}{s_1} = \frac{s_2 \cdot K}{s_1} \quad \Rightarrow C_1 = \frac{s_2 \cdot K}{s_1} \Big/ \left(1 - \frac{s_2}{s_1}\right) = \frac{s_2 \cdot K}{s_1 - s_2}$$

$$\Rightarrow C_2 = -K - \frac{s_2 \cdot K}{s_1 - s_2}$$

$$\Rightarrow y(t) = K + K\frac{s_2}{s_1 - s_2}e^{s_1 t} - K\left(1 + \frac{s_2}{s_1 - s_2}\right)e^{s_2 t} = K\left[1 + \frac{s_2}{s_1 - s_2}e^{s_1 t} - \frac{s_1}{s_1 - s_2}e^{s_2 t}\right]$$

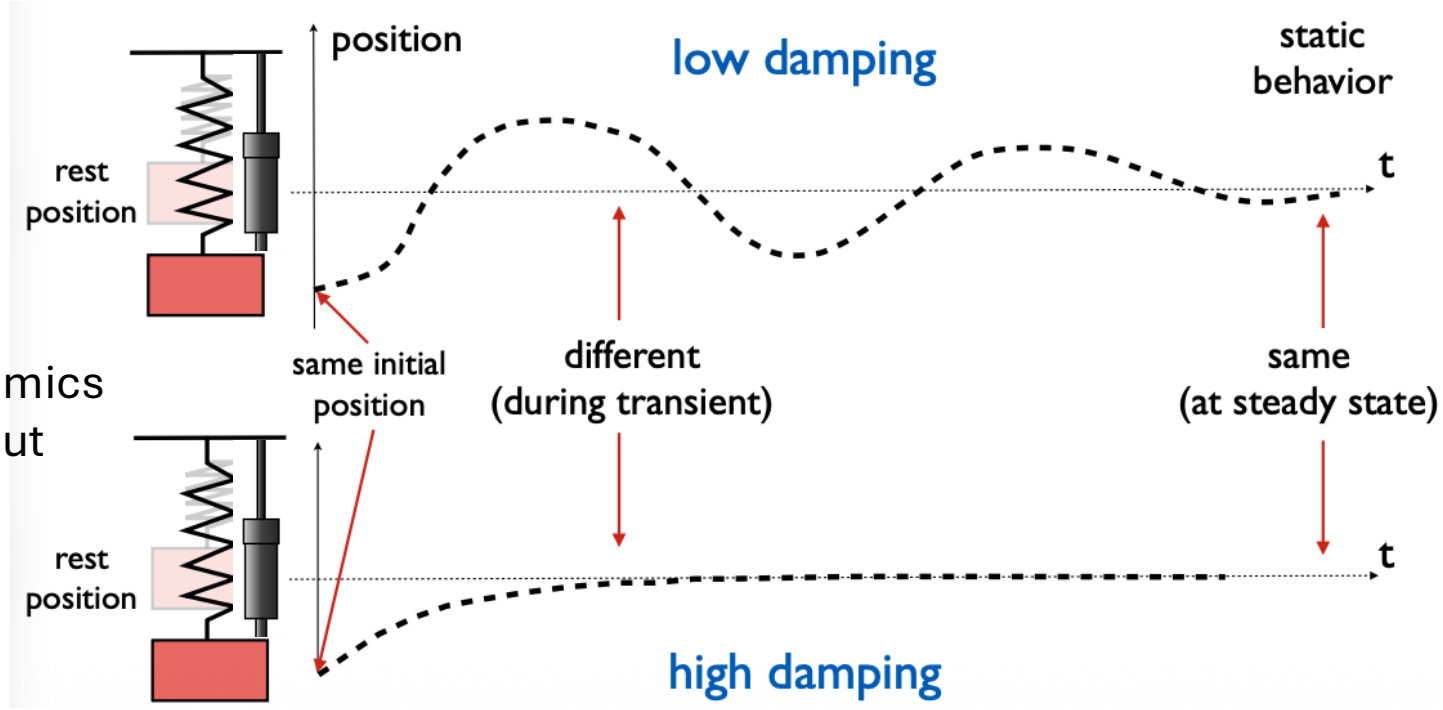
Example: Mass-Spring-Damper

$$m\ddot{x} + \mu\dot{x} + kx = F(t)$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\zeta = \frac{\mu}{2m\omega_n} = \frac{\mu}{2\sqrt{km}}$$

impulse:
same dynamics
as step input



At steady state:

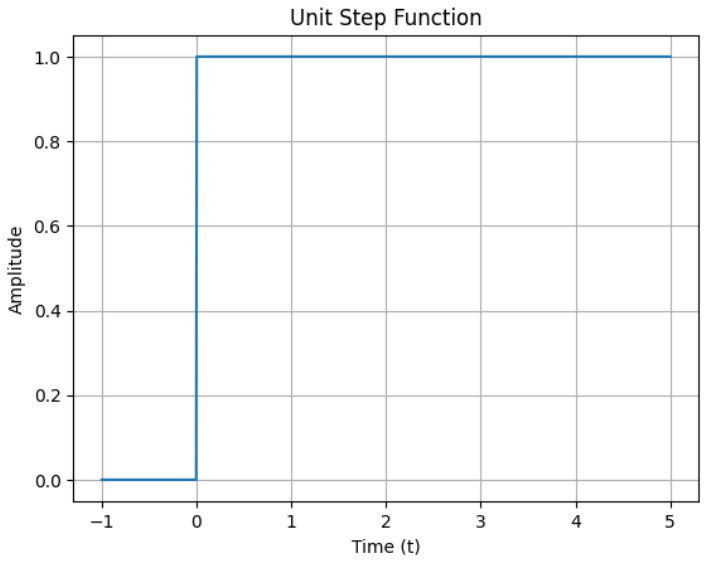
$$\ddot{x} = 0, \dot{x} = 0 \quad \Rightarrow \quad kx(\infty) = F(\infty)$$

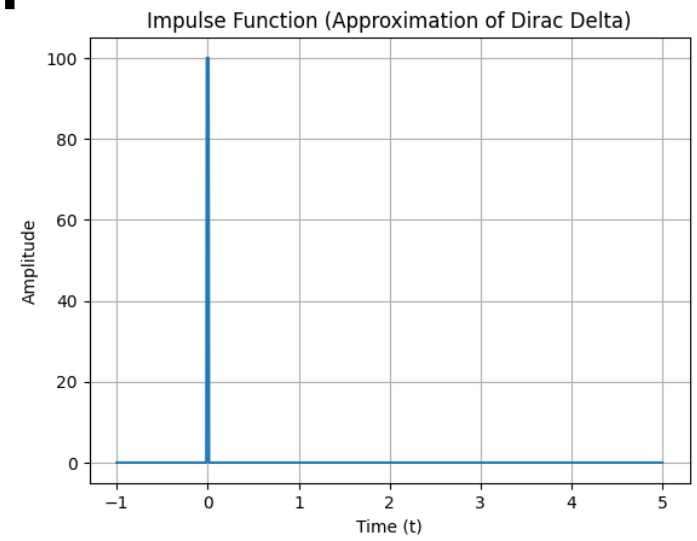
$$K = \frac{x(\infty)}{F(\infty)} = \frac{1}{k}$$

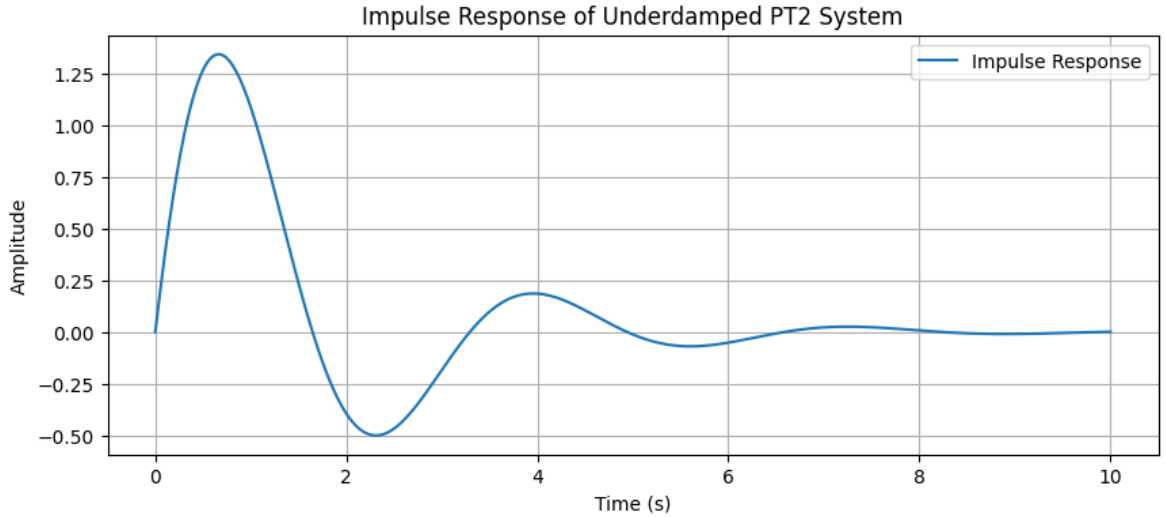
(for step input)

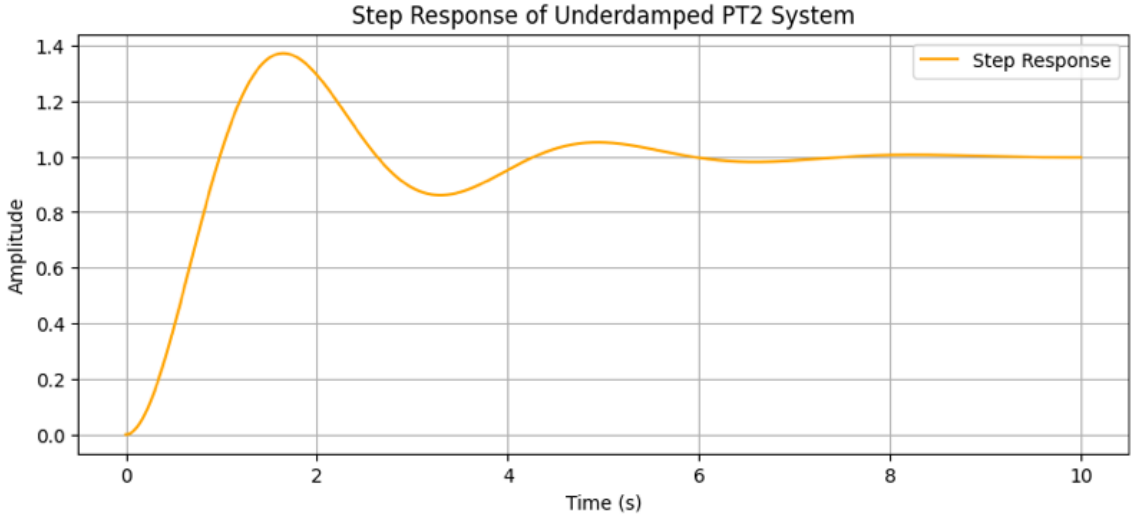
without damping ($\zeta = 0$): $y(t) = A \cdot \cos(\omega_n t + \phi)$

Step and Impulse Response

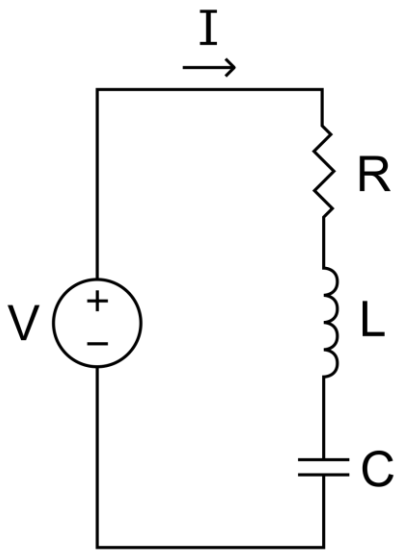








Example: RLC Series Circuit



step voltage input

$$LC\ddot{v}_C + RC\dot{v}_C + v_C = u(t)$$

output: capacitor voltage

$$\omega_n = \sqrt{\frac{1}{LC}}, \quad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}, \quad K = 1$$

- Without R: undamped
- An RLC circuit is a PT2 system whose step response is governed by natural frequency and damping, exactly like a mass–spring–damper system.
- Frequency response (resonance) describes sinusoidal forcing. Here we focus on step response and transient behavior.

Linear Time-Invariant (LTI) Systems

What do PT1 and PT2 have in common?

- Linear equations
- Constant parameters
- Superposition holds (scaling and additivity)
- Time shifts do not change behavior

$$u(t) \rightarrow y(t)$$
$$\Rightarrow a \cdot u(t) \rightarrow a \cdot y(t)$$

$$u_1(t) \rightarrow y_1(t),\ u_2(t) \rightarrow y_2(t)$$
$$\Rightarrow u_1(t) + u_2(t) \rightarrow y_1(t) + y_2(t)$$

→ LTI systems

- Governed by linear differential equations
- Parameters do not depend on time

Foundation of classical control theory

Dead Time

Time delay between input and output: $y(t) = u(t - \tau_d)$

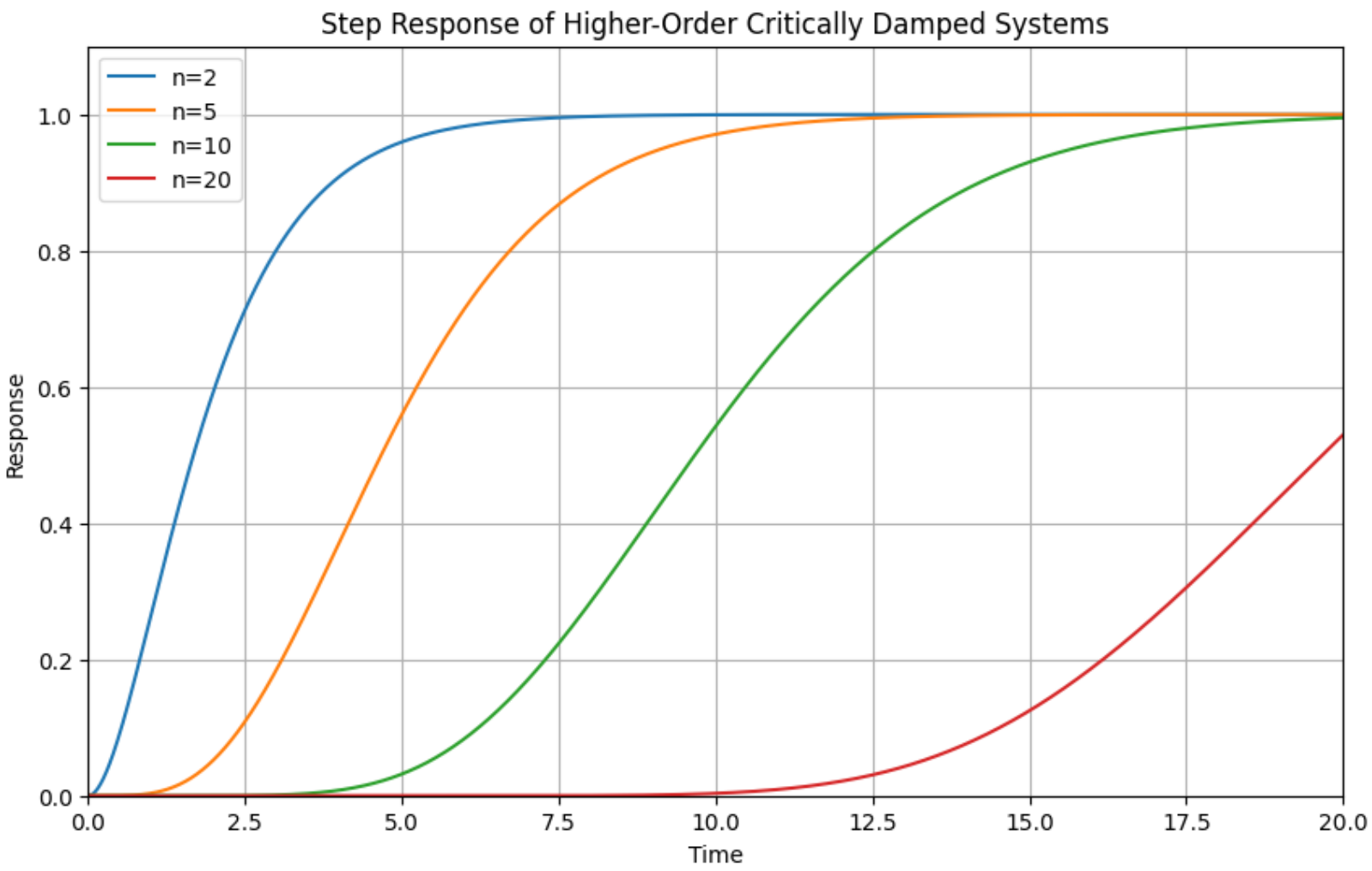
Common in processes and networks

- Transport phenomena
- Measurement delays
- Computation delays

Makes control more difficult

- Degrades stability margins
- Limits achievable performance

Step Responses of Higher-Order Systems



The higher the order, the more the step response looks like a delayed step, even though there's no actual dead time.

Outlook: Frequency Response

So far, we tested systems with steps.

What if the input oscillates?

- slow sinusoid → output follows
- fast sinusoid → output attenuated / phase-shifted

Summary: From Physics to Model

- Identify inputs and outputs
- Apply physical laws
- Simplify assumptions
- Validate against measurements